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Knowledge graph-based cognitive learning with multi-fact reasoning

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ABSTRACT

An intelligent education system is a technology-driven comprehensive education system that is designed to support personalized learning by analyzing interactions between students, learning tasks, and knowledge concepts. Cognitive diagnosis (CD) in an intelligent educational system aims to diagnose the cognitive state of a student from historical response data, and further predict the future performance of the student on the exercise. However, existing CD models still suffer from the following issues: (1) most models only consider the score interactions of student-exercise and exercise-concept, which are difficult to capture the complex interactions among student-exercise-concept; and (2) most models neglect the higher-order interactions among student-exercise-concept. To address these issues, we propose a Cognitive Diagnosis framework based on Signed Knowledge Graph with multi-fact reasoning (CD-SKG). Specifically, we model three core elements in each student response log–student, exercise, and relevant knowledge concept–as individual facts. Since each response has two possible outcomes, we further define these facts as signed facts to encode response valence. Each signed fact inherently contains three distinct entity types, enabling systematic exploration of higher-order relationships within it across three levels, which aligns with our tripartite interaction logic. To distinguish the response impact on student performance, we employ a dual-view hypergraph convolutional network and learn the features of different responses on a signed cognitive hypergraph. In addition, we aim to reveal the different higher-order relations in each response record, which are analyzed at two diagnosis levels by integrating the higher-order information among the three cognitive factors. Finally, we conduct extensive experiments on four real datasets and achieve optimal performance in comparison with eight state-of-the-art CD models. Our code is publicly available at: https://github.com/jianruichen/CD-SKG.

1. Introduction

Cognitive diagnosis (CD) in intelligent education systems is to learn the cognitive state of students by analyzing their response data, and then provide them with personalized learning guidance, such as learning assessments (Bi et al., 2020) and resource recommendations (Guan et al., 2025a, 2023). The personalized approach to education better meets the learning needs of students and help them to master knowledge concepts more effectively (Guan et al., 2025b). CD combines knowledge and technology from a variety of fields (Burns et al., 2014), including pedagogy, cognitive science and information technology, and has been gradually developed and refined. With the continuous advancement of artificial intelligence and big data technology, cognitive diagnostic model (CDM) has become increasingly accurate and reliable in providing powerful support for education and teaching (Chen et al., 2022; Li et al., 2022)

There are two main types of existing CDMs. One type is the classical CDM, which is mainly based on psychological (Templin & Henson, 2006) and cognitive science theories (Thagard, 2013) to infer the cognitive states and learning styles of students by analyzing their learning behaviors and responses. Classical test theory (DeVellis, 2006) is one of the earliest testing theories in psychometrics (Raykov & Marcoulides, 2011), which mainly focuses on the statistical characteristics of test scores and analyzes the measurement results of students. However, early methods focus only on overall scores and consider that test scores consist of true scores and measurement error. Later, item response theory (IRT) uses parameter estimation method to evaluate the potential student ability and exercise characteristic based on the measured characteristics of each exercise in the test (Embretson & Reise, 2013). Then, multi-dimensional IRT maps student abilities and exercise features on multiple dimensions using a logistic-like function (Ackerman et al., 2003). To learn the

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specific knowledge concept mastery on student performance, student response data is combined with the exercise-concept correlation matrix to indicate whether each student masters a knowledge concept with a binary indicator variable (De La Torre, 2009). Nevertheless, classical methods still fail to express the mastery of knowledge concepts in terms of continuous values.

Therefore, another type of CDM has emerged that leverages deep learning to model the features of students and exercises, enabling more precise assessment of students' cognitive states. Neural CDM learns a student-exercise interaction function to predict the mastery of each knowledge concept through the neural network (Wang et al., 2020). To mine the additional information in complex interactions, such as the relations between knowledge concepts. The extended neural CDM with knowledge correlation factors introduces implicit associations between knowledge concepts to improve diagnostic results (Wang et al., 2022). Then, the knowledge-sensed CD enriches the features of students and exercises by exploring higher-order relations between knowledge concepts (Ma et al., 2022). And the relations between exercises and knowledge concepts also affect the mastery performance of students. A neural network is used to combine the importance of the knowledge concepts in exercises through an attention mechanism and the intrinsic connection of knowledge concepts (Gao et al., 2022). To make the results more convincing, the interpretable CD not only makes full use of the connections between knowledge concepts and the quantitative relations between exercises and knowledge concepts, it also considers the effects of the parameters of slip and guess on cognitive states (Qi et al., 2023). In addition, the features of the student and the textual content in exercises have an important influence on the cognitive state. For example, global and local neural CD combines the complementary advantages of CD in student performance prediction tasks to capture the global static and local dynamic cognitive states of students (Su et al., 2024). Then, the cross-modal neural CD improves cognitive diagnostic performance by exploring rich visual and textual content in cross-modal exercises to learn distinguishable features of the exercises (Song et al., 2023). Recently unified uncertainty estimation for CDM finds that the uncertainty factor in measuring the cognitive state of students also provides valid information due to the weak connection between the model and the data (Wang et al., 2024a). However, the above models do not simultaneously improve the interpretability and generalization of CD. Thus symbolic CDM utilizes symbolic tree to interpret complex student-exercise interaction functions and utilizes gradient-based optimization methods to efficiently learn student and exercise parameters (Shen et al., 2024). Furthermore, to integrate the inter-layer information (e.g., student-exercise, exercise-concept) and intra-layer information (e.g., concept-concept), relation map driven CD models complex interactions with a student-exercise-concept multi-layer relation graph (Gao et al., 2021). Graph-based CDM directly discovers interactions between different entities through heterogeneous cognitive graph (Su et al., 2022). In addition, Hypergraph captures more complex relations between entities, which is popular in mining the cognitive learning (Li et al., 2024; Wu & Ling, 2023a; Zhang, 2008). Dual graph ensemble learning method reveals the relevant knowledge concepts as students interact with the knowledge concepts involved in the exercises by constructing the conceptual association hypergraph (Cui et al., 2024).

These models make use of primitive binary relations between different entities or the same entities without recognizing the deeper structure of relations between multiple entities, such as higher-order structures. As shown in Fig. 1(A), existing models use binary relations of student-exercise and exercise-concept to obtain the cognitive state of *Lucy* through CDM and then predict the score on future exercise q_3. However, they neglect the higher-order relations that exist among the three types of entities which possibly impact the cognitive state of *Lucy*. Fig. 1(c) illustrates the higher-order relations between Lucy's answered exercises and the relevant knowledge concepts, such as the second-order relation between c_2 and c_5, and the third-order relation between Lucy and c_2 and c_5, etc. To effectively answer exercise q_3, Lucy must not only understand the five knowledge concepts but also their interrelationships and influences to build a comprehensive knowledge system. Higher-order relations between various knowledge concepts and Lucy offer valuable insights for predicting her score on the exercise q_3. To mine more potential higher-order relations between different entities within the same response log, we apply knowledge graph (KG) to represent the complex relations among student-exercise-concept.

KG is a graphical model for describing the relations between entities, which usually consists of a set of entities and the relations between them (Abu-Salih & Alotaibi, 2024). However there are more complex relations in real-world, and traditional KG only deal with simple binary relations. The emergence of knowledge hypergraph (KHG) provides a more accurate depiction of multiple relations between entities, which means that one relation involves two or more entities (Fatemi et al., 2019). In the cognitive diagnosis task of an intelligent educational system, we explore more complex relations between different entities using basic binary relations. As shown in Fig. 1(c), we explore a variety of higher-order structures among student-exercise-concept through the original binary relations. The higher-order cognitive state of *Lucy* is then obtained based on these higher-order structural information for prediction exercise q_3.

To further analyze the complex higher-order relations between the multiple entities and to depict them more effectively, we propose a Cognitive Diagnosis framework based on Signed Knowledge Graph with multi-fact reasoning (CD-SKG). Signed knowledge hypergraphs reflect students' positive/negative cognitive differences, which support cognitive analysis. The main contributions are summarized as follows:

- How to explore the complex relations among these three types of entities? We formalize multiple entity interactions through a signed cognitive knowledge hypergraph, where student-exercise-concept relations are encoded by signed hyperedges derived from response logs. Each log's label (correct/incorrect) determines hyperedge polarity (positive/negative).
- How to distinguish the impact of different response types on student performance? We employ a dual-channel hypergraph convolutional network to capture the distinctive features of various response types of a signed cognitive hypergraph. This approach enables us to separately model and analyze the unique influences exerted by different response types.
- How to reveal the different higher-order relations in each response log? We integrate higher-order structural information from three cognitive factors at two diagnosis levels. Hierarchical attention jointly encodes local cognitive states and global knowledge topology, disentangling complex interdependencies beyond pairwise models.

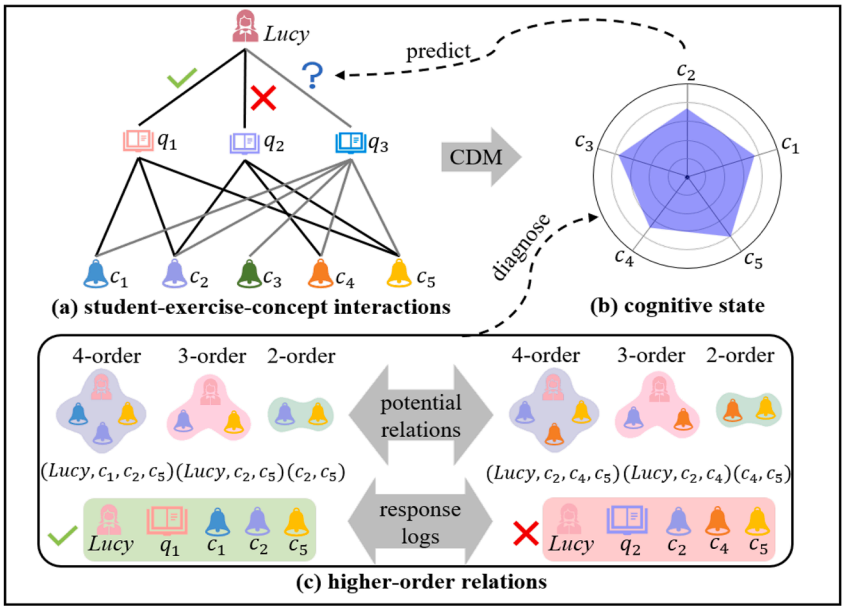
The rest of the paper is organized as follows. Section 2 presents the related work, Section 3 introduces the symbol definitions and task descriptions, Section 4 describes the proposed CD-SKG method in detail, Section 5 displays the experimental results and the corresponding analysis, and Section 6 makes a conclusion.

2. Related work

In this section, we review the related work on hypergraph-based models and knowledge graph relation prediction.

2.1. Hypergraph-based models

Hypergraph expresses higher-order relations and many-to-many connections (He et al., 2025), which provides a deeper understanding of the cognitive states of students in a more precise representation. The construction of self-attentive heterogeneous hypergraph networks helps to capture heterogeneous and higher-order relations between exercises as well as to perceive more important higher-order structures (Wu & Ling, 2023b). Meantime, hypergraph is used to model the dynamic higher-order relations formed in exercise-answer sequences, which is then used



Exercise	Exercise Content	Concepts
q_1	$5-\frac{2}{3}=?$	c_1, c_2, c_5
q_2	$7\frac{1}{4}-\frac{3}{4}=?$	c_2, c_4, c_5
q_3	$8-5\frac{5}{6}=?$	c_1, c_2, c_3, c_4, c_5

Concept	Concept Content
c_1	Convert integer to fraction
c_2	Separate integer from fraction
c_3	Simplify before subtracting
c_4	Borrow from integer part
c_5	Subtract numerators

(B) The content of exercises and related concepts

Fig. 1. The illustrative examples of (A) the CD system based on student *Lucy*; (B) the content of exercises and related concepts.

for learning student embeddings (Ye & Shan, 2023). Then, the dual-channel adaptive scale hypergraph encoders with cross-view contrastive learning model captures the unpaired higher-order relations between responses through a hypergraph distillation component (Li et al., 2024). To explore higher-order relations between students, multi-view hypergraph neural network is proposed to aggregate higher-order information from multi-behavioural data on neighbours as a way to predict student performance (Li et al., 2022). Subsequently, framelet-based dual hypergraph neural networks generate various hypergraphs efficiently from both low-pass and high-pass views as a way to capture complex higher-order relations between students (Yang et al., 2024). Even if higher-order relations between students or exercises through hypergraphs are widely used for student performance prediction, however various higher-order relations between different entities still need to be further explored.

To better capture the complex interactions among student-exercise-concept, we convert response of a student as a signed fact and construct a signed cognitive hypergraph to explore potential higher-order relations between different entities in a fact.

2.2. Knowledge graph relation prediction

KG relation prediction is mainly used to predict unknown relations between entities. Graph neural network (GNN) plays an important role in relation prediction which effectively captures graph structure information and learns complex relation patterns by iteratively updating node representations (Zhou et al., 2020). In particular, graph convolutional network (GCN) effectively learns complex nonlinear relations and contextual information between entities (Schlichtkrull et al., 2018). Then, hypergraph convolutional network (HGCN) is designed to analyze higher-order connection patterns among multiple entities to predict relations in complex data structures more accurately (Feng et al., 2019). For instance, HGCN-based student knowledge profile model enriches the knowledge concept mastery of students by capturing higher-order information between knowledge concepts (Wu et al., 2024). However, to explore the complex relations within higher-order facts, the relation prediction model based on higher-order KHGs aggregates the higher-order information between entities within the facts (Wang et al., 2024b).

To further mine the higher-order information, we model the interactions in CD as knowledge graph. Differently, we learn the features of student responses on two subgraphs of the signed cognitive hypergraph, respectively.

Table 1
Notations and explanations.

Notations	Explanations
V_s	the set of student entities $V_s=\{s_1,s_2,\ldots,s_{n_s}\}$, and $\lvert V_s\rvert=n_s$.
V_q	the set of exercise entities $V_q=\{q_1,q_2,\ldots,q_{n_q}\}$, and $\lvert V_q\rvert=n_q$.
V_c	the set of knowledge concepts $V_c=\{c_1,c_2,\ldots,c_{n_c}\}$, and $\lvert V_c\rvert=n_c$.
L	the set of response logs between students and exercises.
Q	relations between exercises and knowledge concepts, and $Q\in\mathbb{R}^{n_q\times n_c}$.
$\mathcal{V}$	the set of entities $\mathcal{V}=\{v_1,v_2,\ldots,v_{n_e}\}$, and $\lvert\mathcal{V}\rvert=n_e=n_s+n_q+n_c$.
$\mathcal{E}$	the set of different types edges between entities.
$\mathcal{R}$	the set of relation types, and $n_r=\lvert\mathcal{R}\rvert$ is the number of relation types.
$\mathcal{T}$	the set of facts in the task.
M_{re}	the correlation matrix between entities and relations.
M_e	the adjacency matrix between entities.

3. Preliminaries

In this section, we provide a preliminary introduction to knowledge graph-based cognitive learning. Firstly, we model the student-exercise-concept interactions as KHG. Secondly, we define the signed cognitive hypergraph and the CD task based on it. Additionally, the main symbols and their corresponding descriptions in Table 1.

In the CD surroundings, there existing three types entities, i.e., student entities and its corresponding set $V_s=\{s_1,s_2,\ldots,s_{n_s}\}$, exercise entities and its corresponding set $V_q=\{q_1,q_2,\ldots,q_{n_q}\}$, knowledge concept entities and its corresponding set $V_c=\{c_1,c_2,\ldots,c_{n_c}\}$. The answers of the students on the exercises constitute the response logs $L=(s_i,q_j,r_p)$, where $s_i\in V_s(i=1,2,\ldots,n_s)$ and $n_s=\lvert V_s\rvert$ is the number of students. $q_j\in V_q$ $(j=1,2,\ldots,n_q)$ and $n_q=\lvert V_q\rvert$ is the number of exercises. $r_p(p=0,1)$ mean two relations between students and exercises, where r_0 represents the incorrect response and r_1 represents the correct response. The associations between the exercises and the knowledge concepts constitute the correlation matrix $Q=(\mathrm{q}_{jk})\in\mathbb{R}^{n_q\times n_c}$, where $\mathrm{q}_{jk}\in\{0,1\}$. $\mathrm{q}_{jk}=1$ means the knowledge concept c_k is included in the exercise q_j, otherwise $\mathrm{q}_{jk}=0$. Different involved concepts $c_k\in V_c(k=1,2,\ldots,n_c)$, and $n_c=\lvert V_c\rvert$ is the number of knowledge concepts.

Definition 1 (Multi-fact reasoning). Multi-fact reasoning denotes a specialized reasoning paradigm that operates on knowledge hypergraphs, wherein a single hyperedge connects multiple entities (e.g., students, exercises, knowledge concepts). This structure enables the simultaneous modeling and analysis of complex interactive relationships among these entities. The core objective of this reasoning paradigm is to mine implicit higher-order patterns and structural dependencies by analyzing the concurrent multi-entity associations embedded in each response log.

To directly capture the higher-order structure of interactions among students, exercises and concepts, we regard each response log and the associated knowledge concepts as a fact in KHG, rather than multiple binary relationships between distinct entities in traditional knowledge graphs. Each fact $t = r_p(s_i, q_j, c_1, \dots, c_m)$ means response relation r_p of student entity s_i on exercise entity q_j and concept entites $\{c_1, \dots, c_m\}$, where $\{c_1, \dots, c_m\}$ is the set of knowledge concepts contained in the exercise q_j. However, real-world learning processes and assessment data are often characterized by multi-dimensionality and complexity: a single test item may involve interactions among multiple attributes; a learner's performance on a task may reflect concurrent influences of multiple factors; and relationships between attributes are rarely one-dimensional. These features align closely with the core of multi-fact reasoning –the ability to synthesize multiple concurrent facts to infer hidden patterns or resolve conflicts. Integrating multi-fact reasoning with CD can address key challenges in CD, such as handling attribute interactions, mitigating data ambiguity, and enhancing diagnostic granularity. It is worth noting that in our approach, "multi-fact" refers to the complex interactions involving multiple entity types (students, exercises, individual knowledge concept, or multiple knowledge concepts) coexisting within each response log, rather than the traditional notion of multiple relationships.

Definition 2 (Signed cognitive KHG). We define student-exercise-concept interactions as a signed cognitive KHG $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \mathcal{T})$. In particular, $\mathcal{V} = V_s \bigcup V_q \bigcup V_c$ consists of students, exercises and concepts. $\mathcal{E} = \mathcal{E}_0 \bigcup \mathcal{E}_1 \bigcup \mathcal{E}_2$ is the set of distinct edges between entities, where $\mathcal{E}_0$, $\mathcal{E}_1$ and $\mathcal{E}_2$ denote the set of negative edges between students and exercises (i.e., the incorrect responses), the set of positive edges between students and exercises (i.e., the correct responses) and the set of edges between exercises and knowledge concepts (i.e., the edges in correlation matrix Q), respectively. $\mathcal{R} = \{r_0, r_1\}$ is the set of two types response relations. $\mathcal{T}$ is the set of facts in the system, and each fact $t = r_p(s_i, q_j, c_1, \dots, c_m)$ corresponds by relation r_p student s_i responds to exercise q_j containing concepts $\{c_1, \dots, c_m\}$.

Collectively, the signed knowledge graph with multi-fact reasoning enables us to structure student response data into interpretable signed triples and explore tripartite interactions, i.e., student-excercise-knowledge concept through cross-fact inference. As shown in Fig. 2, *Lucy* responds to exercise q_1 correctly and exercise q_1 contains knowledge concepts $\{c_1, c_2, c_5\}$. We construct a positive fact among the three types of entities: student *Lucy*, exercise q_1, and knowledge concepts c_1, c_2 and c_5. The positive fact is expressed as: $t_1 = r_1(Lucy, q_1, c_1, c_2, c_5)$. Similarly, we construct a negative fact as: $t_2 = r_0(Lucy, q_2, c_2, c_4, c_5)$.

Definition 3 (The task of cognitive diagnosis). Given the signed cognitive KHG $\mathcal{G}$, the goal of cognitive diagnosis in $\mathcal{G}$ is to obtain the cognitive state α_i (i.e., knowledge mastery) of student s_i by predicting the probability of student s_i responsing to exercise q_j. In other words, cognitive diagnosis in KHG is to predict the probability of the target fact $t = r_p(s_i, q_j, c_1, \dots, c_m)$.

In Fig. 2, the task of cognitive diagnosis is to obtain the mastery of each knowledge concept through the responses to the answered exercises (i.e., q_1 and q_2) of *Lucy* and then to predict whether *Lucy* responds q_3 (related to concepts c_1, c_2, c_3, c_4 and c_5) correctly. That is to say, the task is to predict the probability of signed fact $(Lucy, q_3, c_1, c_2, c_3, c_4, c_5)$.

4. Proposed method

In this section, we first introduce the overall framework of CD-SKG. Next, we introduce the construction of signed cognitive KHG in Section 4.1. Then, we design the dual-view HGCN to learn the embeddings of different entities in Section 4.2. Finally, we introduce how to diagnose and predict via three cognitive factors in Section 4.3.

Fig. 3 illustrates the overall framework of our proposed CD-SKG, which contains the following main aspects. In Fig. 3(A), we construct a signed cognitive hypergraph through multi-factual information, shown in Section 4.1. In Fig. 3(B), we learn the features of relations and entities on the signed cognitive hypergraph through the dual-view HGCN in Section 4.2. Then, we obtain the features of each entity in the fact used to obtain higher-order features. In Fig. 3(C), we explore higher-order relations under two diagnosis levels for each signed fact. Finally, we combine three cognitive factors to predict the probability of a signed fact.

4.1. Signed cognitive hypergraph construction

Traditional heterogeneous graphs are limited to modeling binary relationships, such as student-exercise or exercise-knowledge concept associations. In contrast, our signed cognitive hypergraph directly encodes the student-exercise-knowledge concept triplet as an atomic fact, thereby enabling a more natural representation of complex higher-order interactions among these entities. This paper only explores the complex relations among three entity types without considering the relations within the same entity type. For the student response logs, there are two types of relations between students and exercises: correct and incorrect responses. According to the monotonicity assumption theory (Rosenbaum, 1984), if a student master all concepts in the exercise, it is more likely that the student correctly answers the exercise. Therefore, the different responses of students reflect their different cognitive states. To differentiate the different student responses, we construct two subgraphs based on the signed cognitive hypergraph: positive cognitive hypergraph and negative cognitive hypergraph. By decomposing positive and negative responses into two independent hypergraphs, our model not only learns distinct feature representations for each response type but also avoids interference between positive and negative cognitive features.

For the positive cognitive hypergraph, we use the edges $\mathcal{E}_1$ and $\mathcal{E}_2$ to construct the positive correlation matrix of student-exercise and exercise-concept $M_e^+ \in \mathbb{R}^{n_e \times n_e}$, shown in Eqs. (1) and (2).

$$M_e^+ = \begin{bmatrix} O & M_{sq}^+ & O \\ M_{qs}^+ & O & M_{qc} \\ O & M_{cq} & O \end{bmatrix}_{n_e \times n_e}, \tag{1}$$

$$\underset{(n_s \times n_q)}{M_{sq}^+}(i,j) = \begin{cases} 1, & if\ (s_i, q_j) \in \mathcal{E}_1; \\ 0, & else, \end{cases} \qquad \underset{(n_q \times n_c)}{M_{qc}}(j,k) = \begin{cases} 1, & if\ (q_j, c_k) \in \mathcal{E}_2; \\ 0, & else. \end{cases} \tag{2}$$

Here, $n_e = n_s + n_q + n_c$ is the number of entities in the network. M_{sq}^+ represents the correlation block-matrix of correct responses between students and exercises. M_{qc} represents the contained block-matrix between exercises and the knowledge concepts. M_{qs}^+ and M_{cq} represent the transpose block-matrices of M_{sq}^+ and M_{qc}, respectively. O is a full-zero matrix with corresponding dimensions. $(s_i, q_j) \in \mathcal{E}_1$ means student s_i answers exercise q_j correctly and $(q_j, c_k) \in \mathcal{E}_2$ means exercise q_j contains concept c_k. Each positive fact $(s_i, q_j, c_1, \dots, c_k)$ represents a hyperedge in the positive cognitive hypergraph, which contains the student s_i, the correctly answered exercise q_j and its related concepts $c_1, \dots, c_k$.

However, for the negative cognitive hypergraph, we construct the negative correlation matrix of student-exercise and exercise-concept $M_e^- \in \mathbb{R}^{n_e \times n_e}$, shown in Eqs. (3) and (4). $(s_i, q_j) \in \mathcal{E}_0$ means student s_i answers exercise q_j incorrectly.

$$M_e^- = \begin{bmatrix} O & M_{sq}^- & O \\ M_{qs}^- & O & M_{qc} \\ O & M_{cq} & O \end{bmatrix}_{n_e \times n_e}, \tag{3}$$

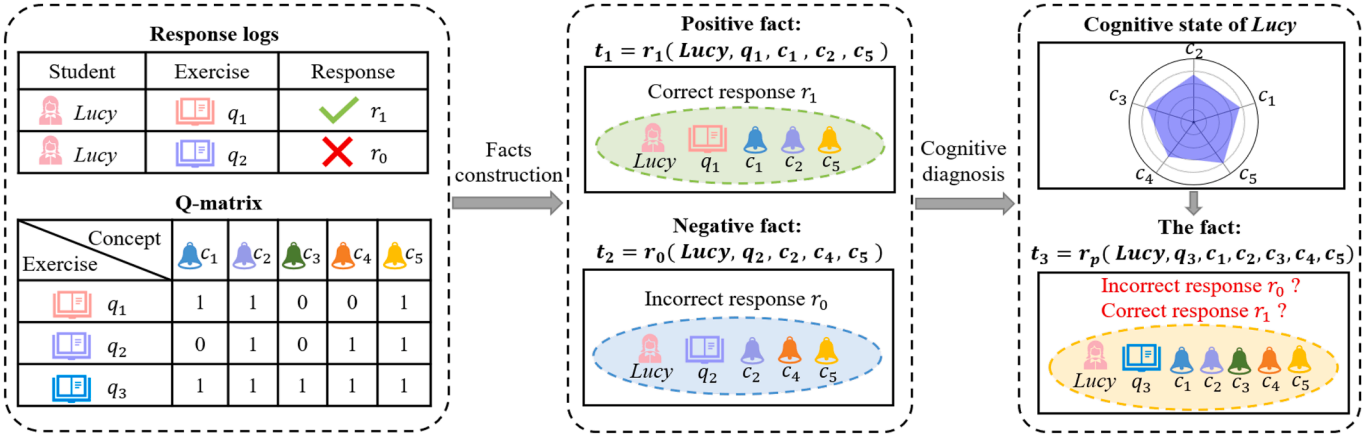


Fig. 2. Illustrative examples of signed cognitive KHG and the task of cognitive diagnosis.

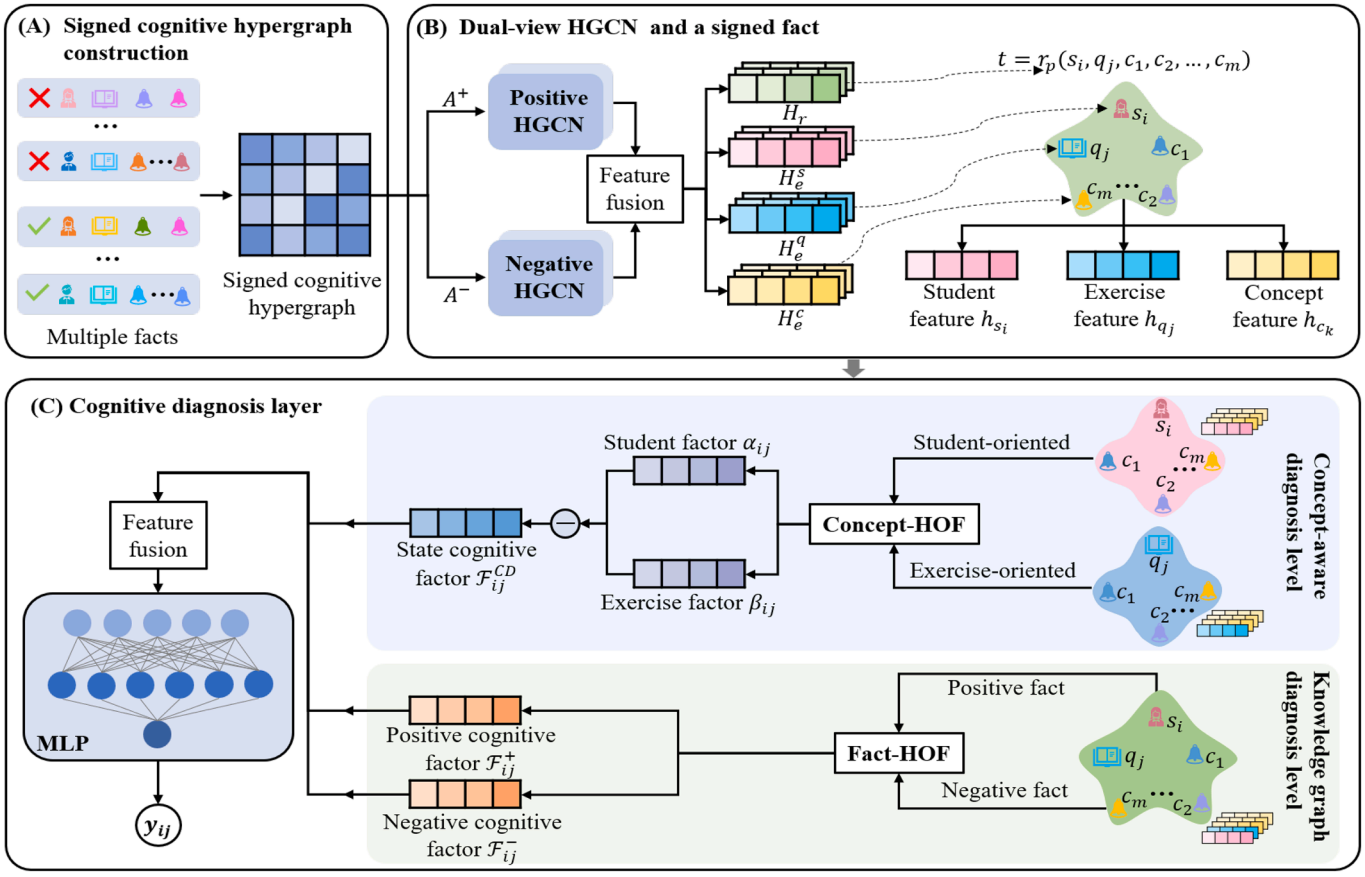


Fig. 3. The overall framework of the proposed CD-SKG.

$$M_{sq}^{-}(i,j) = \begin{cases} 1, & if\ (s_i, q_j) \in \mathcal{E}_0; \\ 0, & else, \end{cases} \tag{4}$$

where M_{sq}^{-} represents the correlation matrix of incorrect responses between students and exercises. M_{qs}^{-} represents the transpose block-matrix of M_{sq}^{-}. M_{qc} and M_{cq} are as mentioned in Eq. (2). Each negative fact $(s_i, q_j, c_1, \ldots, c_k)$ represents a hyperedge in the negative cognitive hypergraph, which contains the student s_i, the incorrectly answered exercise q_j and its related concepts $c_1, \ldots, c_k$.

Then, we construct the correlation matrix $M_{re} \in \mathbb{R}^{n_r \times n_e}$ between relations and entities, here $n_r = |\mathcal{R}|$ is the number of relation types in the network. If fact $t = r_p(s_i, q_j, c_k)$ in the set of facts $\mathcal{T}$, we set:

$$M_{re}(p,i) = M_{re}(p, j+n_s) = M_{re}(p, k+n_s+n_q) = \begin{cases} 1, & if\ r_p(s_i, q_j, c_k) \in \mathcal{T}; \\ 0, & else. \end{cases} \tag{5}$$

Finally, we combine the correlation matrix M_{re} and the adjacency matrices M_e^{+} and M_e^{-} to obtain the cognitive matrices $A^{+} \in \mathbb{R}^{(n_r+n_e)\times(n_r+n_e)}$ of the positive cognitive hypergraph and $A^{-} \in \mathbb{R}^{(n_r+n_e)\times(n_r+n_e)}$ of the negative cognitive hypergraph:

$$A^{+} = \begin{bmatrix} O & M_{re} \\ M_{re}^{T} & M_e^{+} \end{bmatrix}, \tag{6}$$

and

$$A^{-} = \begin{bmatrix} O & M_{re} \\ M_{re}^{T} & M_e^{-} \end{bmatrix}, \tag{7}$$

where $O \in \mathbb{R}^{n_r \times n_r}$ is a full-zero matrix and M_{re}^{T} is the transpose matrix of M_{re}.

4.2. Dual-view hypergraph learning layer

We propose a dual-view HGCN learning model to differentiate feature representations for various student responses. This not only allows for a more comprehensive understanding of response data, but also differentiates different types of responses, which greatly improves the ability to capture complex relations between entities.

Initially, we encode the randomly initialized feature embedding of relations and entities into the d-dimensional hidden space:

$$H^{(0)} = \psi(H_r^{(0)}, H_e^{(0)}), \tag{8}$$

where $H_r^{(0)} \in \mathbb{R}^{n_r \times d}$ and $H_e^{(0)} \in \mathbb{R}^{n_e \times d}$ are the initialized features of relations and entities, respectively. $\psi(\cdot)$ represents concatenation operation.

Subsequently, dual-view HGCN is used to learn different features in signed cognitive hypergraph. The first view is positive HGCN on the positive cognitive subgraph, which focuses on learning the features of student correct responses. The second view is negative HGCN on the negative cognitive subgraph, which focuses on learning the features of student incorrect responses.

Positive HGCN (P-HGCN). In the first view, we learn the features of relations and entities in the positive cognitive subgraph via HGCN:

$$H_+^{(l+1)} = \sigma_1(D_+^{-\frac{1}{2}} A^+ D_+^{-\frac{1}{2}} H_+^{(l)} W_+^{(l)}), \tag{9}$$

where $H_+^{(l)} \in \mathbb{R}^{(n_r+n_e)\times d}$ is the embedding matrix of the lth layer of all relations and entities in P-HGCN. A^+ is the adjacency matrix of the positive cognitive subgraph shown in Eq. (6). $D_+^{-\frac{1}{2}}$ is the normalized degree matrix of the matrix A^+. $W_+^{(l)}$ is the weight matrix of layer l in P-HGCN. σ_1 is the activation function. Furthermore, $H_+^{(0)}$ is the initialized features $H^{(0)}$ shown in Eq. (8).

The embeddings learned by P-HGCN effectively capture the overall information between entities and contextual information. We merge the initial features with the ones obtained from each layer to achieve the final features $H^+ \in \mathbb{R}^{(n_r+n_e)\times d}$:

$$H^+ = \frac{1}{1+\theta^+} \sum_{l=0}^{\theta^+} H_+^{(l)}, \tag{10}$$

where θ^+ is the number of layers of P-HGCN and $H_+^{(l)}$ is the features of layer l.

Negative HGCN (N-HGCN). In the second view, we obtain the features of all relations and entities in the negative cognitive subgraph through HGCN:

$$H_-^{(l+1)} = \sigma_1(D_-^{-\frac{1}{2}} A^- D_-^{-\frac{1}{2}} H_-^{(l)} W_-^{(l)}), \tag{11}$$

where $H_-^{(l)} \in \mathbb{R}^{(n_r+n_e)\times d}$ is the lth feature matrix of all relations and entities in N-HGCN. A^- is adjacency matrix of the negative cognitive subgraph shown in Eq. (7). $D_-^{-\frac{1}{2}}$ is the normalized degree matrix of the matrix A^-. $W_-^{(l)}$ is the weight matrix of layer l in N-HGCN. σ_1 is the activation function. Moreover, $H_-^{(0)}$ is the initialized feature $H^{(0)}$ shown in Eq. (8).

Similar to the first view, we obtain the final embedding matrix $H^- \in \mathbb{R}^{(n_r+n_e)\times d}$ of the N-HGCN:

$$H^- = \frac{1}{1+\theta^-} \sum_{l=0}^{\theta^-} H_-^{(l)}, \tag{12}$$

where θ^- is the number of layers of N-HGCN and $H_-^{(l)}$ is the features of layer l.

Positive and negative feature fusion. To capture the cognitive states with correct and incorrect responses, we generate feature fusion matrix $H \in \mathbb{R}^{(n_r+n_e)\times d}$ of all relations and entities based on the comprehensive network perspective:

$$H = g(H^+, H^-), \tag{13}$$

where H^+ and H^- are the features obtained from two cognitive subgraphs, respectively. $g(\cdot)$ represents a feature fusion operation, with detailed analysis in Section 5.3.

After that, we obtain the corresponding features for all different types of entities:

$$H = [\underbrace{H_1, \ldots, H_{n_r}}_{H_r}, \underbrace{H_1, \ldots, H_{n_s}}_{H_e^s}, \underbrace{H_1, \ldots, H_{n_q}}_{H_e^q}, \underbrace{H_1, \ldots, H_{n_c}}_{H_e^c}], \tag{14}$$

where $H_r \in \mathbb{R}^{n_r \times d}$, $H_e^s \in \mathbb{R}^{n_s \times d}$, $H_e^q \in \mathbb{R}^{n_q \times d}$ and $H_e^c \in \mathbb{R}^{n_c \times d}$ are the features of relations, students, exercises and knowledge concepts from dual-view HGCN, respectively. n_r, n_s, n_q and n_c are the number of relations and corresponding entities respectively.

4.3. Cognitive diagnosis layer

After obtaining all the features for relations and entities, we further predict the probability of the target fact $t = r_p(s_i, q_j, c_1, \ldots, c_m)$ in $\mathcal{T}$. Inspired by that the same entity plays different roles in different facts (Wang et al., 2024b), we employ a swapping function $S(\cdot)$ that incorporates embeddings of varying dimensions and randomly swaps them to extract features of different entities. We acquire the feature of each entity in fact t:

$$(h_{r_p}, h_{s_i}, h_{q_j}, h_{c_k}) = S(H_r(p), H_e^s(i), H_e^q(j), H_e^c(k)), \tag{15}$$

where $h_{r_p} \in \mathbb{R}^d$, $h_{s_i} \in \mathbb{R}^d$, $h_{q_j} \in \mathbb{R}^d$ and $h_{c_k} \in \mathbb{R}^d$. $c_k \in \{c_1, \ldots, c_m\}$ is the set of related concepts in the fact t.

Unlike conventional cognitive diagnosis methods, we comprehensively analyze the cognitive factors associated with students' responses by adopting two distinct diagnosis levels grounded in each signed fact. First, the concept-aware diagnosis level considers the influence of various higher-order structures among different knowledge concepts within the fact on student factor and exercise factor. Second, the knowledge graph diagnosis level explores the influence of various higher-order information among entities within the fact on correct response factor and incorrect response factor, starting from two possible response outcomes (i.e., correct response and incorrect response). Finally, we synthesize multifaceted factors with higher-order information obtained from both diagnostic levels to predict the performance of student s_i in fact t.

4.3.1. Concept-aware diagnosis level

For the concept-aware diagnosis level, we obtain the state cognitive factor $\mathcal{F}_{ij}^{CD}$ of student s_i on exercise q_j by higher-order feature fusion in fact t. The state cognitive factor $\mathcal{F}_{ij}^{CD}$ consists of two types of factors: the student factor and the exercise factor.

Students' answers to specific exercises are influenced by the interaction between multiple knowledge concepts within that exercise. For example, when facing a exercise about *trigonometric functions*, students not only need to understand various concepts such as *properties*, *graphs*, and *transformations* of *trigonometric functions*, but also need to flexibly apply one or a combination of these concepts to solve specific exercises. Specifically, the factor of student s_i is obtained from the embedding of student s_i and the m knowledge concepts involved in exercise q_j. For student s_i, there are various u-order structures between student s_i and knowledge concept set C ($C = \{c_1, \ldots, c_m\}$), and each u-order structure has $\binom{m}{u-1}$ substructures. Here, $u \in U$ and $U = \{2, 3, 4\}$ is a set of orders. We fuse the substructure features through a linear transformation function to obtain the u-order structure features, and then integrate the various u-order structural features to obtain the factor of student s_i. In Fig. 4(a), we exemplify the structure between student s_i and various knowledge concepts within fact $t = r_p(s_i, q_j, c_1, c_2, c_3, c_4, c_5)$. As shown in Fig. 4(b), we illustrate the various higher-order substructures. It can be observed that, the second-order structure has $\binom{5}{1}$ substructures, the third-order structure has $\binom{5}{2}$ substructures, and the fourth-order structure has $\binom{5}{3}$ substructures. We fuse the different structural features in each order by means of a linear transformation function, e.g., the second-order features between s_i and knowledge concepts are fused into a_2 through $f_{s_i}^{<2>}$. Subsequently, all high-order structural features are added together, and then the cognitive state of student s_i with high-order features is obtained through the linear transformation function f_{s_i} and activation function σ_2. As shown in Fig. 4(b), $\alpha_{ij} = \sigma_2(f_{s_i}(a_1 + a_2 + a_3))$. Consequently, the u-order structure feature and the higher-order feature of the student are denoted as:

$$a_u = f_{s_i}^{<u>}(\psi(\mathcal{P}(h_{s_i}, h_{c_{k_2}}, h_{c_{k_3}}, \ldots, h_{c_{k_u}}))), \tag{16}$$

and

$$\alpha_{ij} = \sigma_2(f_{s_i}(\Sigma_{u\in U} a_u)), \tag{17}$$

where a_u is the u-order structure feature embedding, $f_{s_i}^{<u>}(\cdot)$ is the u-order linear transformation function that integrates substructures into a space to enhance the correlation between different substructures, $\psi(\cdot)$ is the concatenation operation, and $\mathcal{P}(\cdot)$ is the element-wise product. α_{ij} is a higher-order cognitive state that incorporates all higher-order feature embeddings within fact t, $f_{s_i}(\cdot)$ represents the linear transformation function that maps embedding dimensions into the space of knowledge concepts, and σ_2 is the activation function.

A exercise often encompasses multiple knowledge concepts, and their combinations influence the exercise's complexity. For instance, trigonometric function graph exercises typically focus on the relationship between *angles* and *trigonometric values*. However, considering additional features like *symmetry* and *periodicity* can lead to more efficient solutions. Similarly, we obtain the factor of the exercise q_j in the same way:

$$b_u = f_{q_j}^{<u>}(\psi(\mathcal{P}(h_{q_j}, h_{c_{k_2}}, h_{c_{k_3}}, \ldots, h_{c_{k_u}}))), \tag{18}$$

$$\beta_{ij} = \sigma_2(f_{q_j}(\Sigma_{u\in U} b_u)), \tag{19}$$

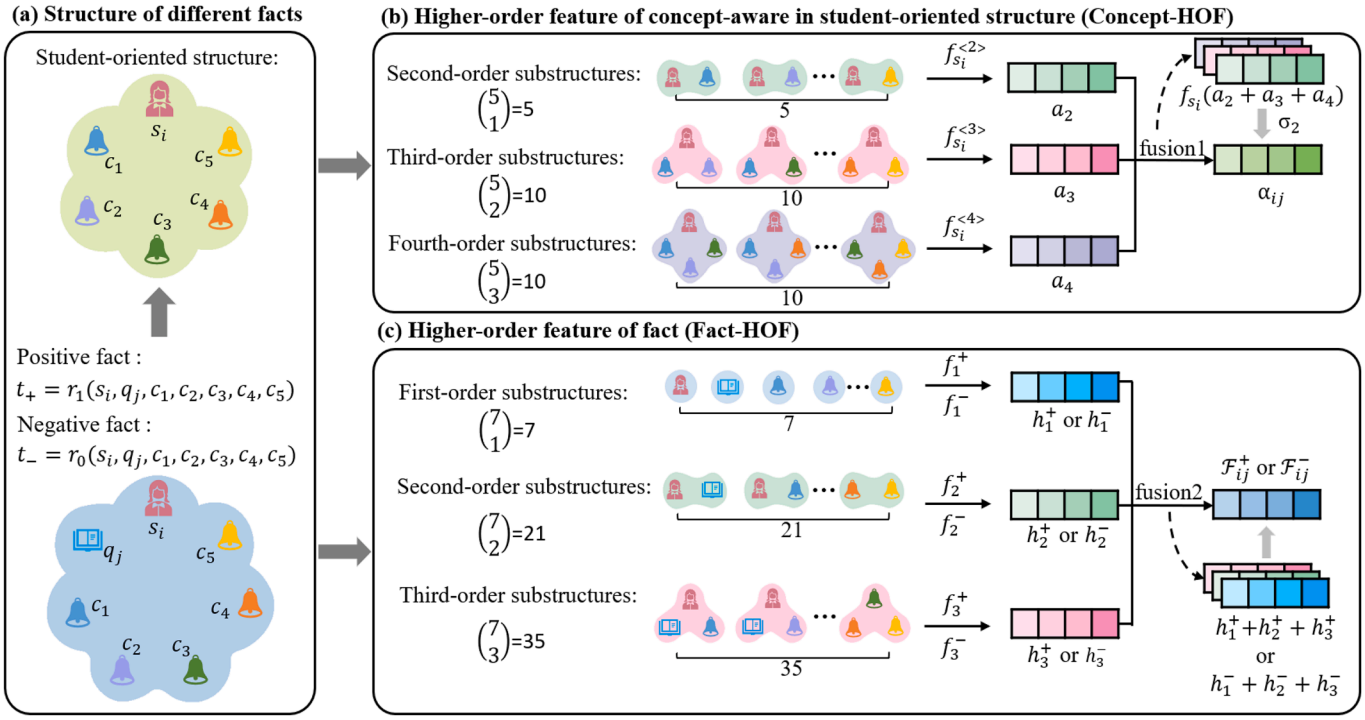


Fig. 4. Illustrative examples of (a) the fact $t = r_p(s_i, q_j, c_1, c_2, c_3, c_4, c_5)$ and student-oriented structure between student and knowledge concepts; (b) the higher-order structure between s_i and knowledge concepts within the fact t and the fusion process of higher-order substructure feature embeddings; (c) the higher-order structure between various entities within the fact t and the fusion process of higher-order substructure feature embeddings.

where b_u is the u-order structure feature embedding, $f_{q_j}^{<u>}(\cdot)$ is the u-order linear transformation function, $\psi(\cdot)$ is the concatenation operation, and $\mathcal{P}(\cdot)$ is the element-wise product in Eq. (18). β_{ij} is a higher-order exercise factor embedding that incorporates all higher-order structural feature embeddings, $f_{q_j}(\cdot)$ represents the linear transformation function that maps embedding dimensions into the space of knowledge concepts in Eq. (19). σ_2 is the activation function.

Subsequently, we combine the factors of student s_i and exercise q_j to gain state cognitive factor $\mathcal{F}_{ij}^{CD}$ for predicting student performance at the cognitive diagnosis level:

$$\mathcal{F}_{ij}^{CD} = \alpha_{ij} - \beta_{ij}. \tag{20}$$

4.3.2. Knowledge graph diagnosis level

At the level of knowledge graph diagnosis, we focus on the two potential relations (i.e., r_0 and r_1) within fact t. For the existence of two possible relations, we gain the two possible facts such as negative fact $t_- = r_0(s_i, q_j, c_1, \ldots, c_m)$ and positive fact $t_+ = r_1(s_i, q_j, c_1, \ldots, c_m)$. As shown in Fig. 4(a), fact $t = r_p(s_i, q_j, c_1, c_2, c_3, c_4, c_5)$ have two possible facts $t_- = r_0(s_i, q_j, c_1, c_2, c_3, c_4, c_5)$ and $t_+ = r_1(s_i, q_j, c_1, c_2, c_3, c_4, c_5)$.

Different from the concept-aware diagnosis level, the higher-order structure at this level exists not only between the student and knowledge concepts or between the exercise and knowledge concepts but also among three different types of entities. Different relations allow for learning richer information about the feature in fact. In addition, to ensure the richness of the information in the higher-order facts, we also take into account the first-order primitive features of all the entities, as shown in Fig. 4(c). Specifically, we first enumerate all possible higher-order structures between different types of entities. Next, multiply the higher-order structural features and the relation features within each order based on the two potential facts t_- and t_+. Then concatenate the resulting features and use the linear transformation function f_z^- or f_z^+ to obtain the higher-order features for each order, such as f_2^- and f_2^+ in Fig. 4(c). Finally, add all the higher-order structural features from the two potential facts t_- and t_+ to obtain the positive cognitive factor $\mathcal{F}_{ij}^-$ and negative cognitive factor $\mathcal{F}_{ij}^+$.

$$\mathcal{F}_{ij}^- = \Sigma_{z\in Z}(f_z^-(\psi(\mathcal{P}(h_{r_0}, h_{e_{a_1}}, h_{e_{a_2}}, \ldots, h_{e_{a_z}})))), \tag{21}$$

$$\mathcal{F}_{ij}^+ = \Sigma_{z\in Z}(f_z^+(\psi(\mathcal{P}(h_{r_1}, h_{e_{a_1}}, h_{e_{a_2}}, \ldots, h_{e_{a_z}})))), \tag{22}$$

where entity e_a is one of the three types of entities in fact t and $Z = \{1, 2, 3\}$ is a set of orders. For each z-order structure, there are $\binom{n_t}{z}$ substructures in it, that means there are $\binom{n_t}{z}$ elements multiplied inside $\mathcal{P}(\cdot)$. n_t is the number of the entities in fact t. $\psi(\cdot)$ is the concatenation operation. $f_z^-(\cdot)$ and $f_z^+(\cdot)$ are the integration functions for all substructures in each z-order structure, such as a linear transformation function that integrates all z-order substructure features together.

4.3.3. Score prediction based on cognitive factors

Since the intricate interactions between various cognitive aspects affect student performance, we consider a combination of these three cognitive factors to make a more comprehensive prediction of student scores. The allied cognitive factor is expressed as:

$$\mathcal{F}_{ij} = \psi(\mathcal{F}_{ij}^{CD}, \mathcal{F}_{ij}^-, \mathcal{F}_{ij}^+), \tag{23}$$

where $\psi(\cdot)$ is the concatenation operation. $\mathcal{F}_{ij}^{CD}$, $\mathcal{F}_{ij}^-$ and $\mathcal{F}_{ij}^+$ are the three cognitive factors obtained from Eqs. (20) to (22), respectively.

For fact t, the probability of fact t which means the prediction score of student s_i on exercise q_j is represented as:

$$y_{ij} = \sigma_2(W_3(W_2(W_1\mathcal{F}_{ij} + b_1) + b_2) + b_3), \tag{24}$$

where W_1, W_2, W_3, b_1, b_2, b_3 are the trainable weight matrices and bias terms of the multilayer perceptual machine, and σ_2 is the activation function.

We then apply a cross entropy loss function to optimize the error between the true score r_p and the final prediction score y_{ij} in fact t:

$$\begin{aligned} loss = &-\Sigma_{(s_i,q_j,r_p)\in L}(r_p \log y_{ij}) + (1 - r_p)(\log(1 - y_{ij})) \\ &+ \lambda(\|W_1\|_2 + \|W_2\|_2 + \|W_3\|_2), \end{aligned} \tag{25}$$

where λ is the weight decay that prevents the model overfitting. W_1, W_2 and W_3 are the weight matrices of the multilayer perceptual machine in Eq. (24).

5. Experiments

In this section, we conduct extensive experiments and explore our proposed CD-SKG in terms of the following five exercises:

- **Q1**: How does our model perform compared to existing methods?
- **Q2**: How are the impact of hyperparameters and feature fusion operation of our model?
- **Q3**: How effective are the signed cognitive hypergraph and two diagnosis levels?
- **Q4**: How is the difference in the embedding of different responses of students learnt by our model in the signed cognitive hypergraph?
- **Q5**: How interpretable is our model with higher-order patterns?

Table 2
Properties of four datasets.

Statistics	FrcSub	JunYi	ASSIT2017	ASSIT0910
#Entities	564	1751	3989	19,294
#Students	536	1000	1678	2,380
#exercises	20	712	2210	16,804
#Knowledge Concepts	8	39	101	110
#Maximum concepts	5	1	2	4
#Response logs	10,720	203,945	351,530	257,585
#Positive response logs	5723	138,751	152,111	170,451
#Negative response logs	4997	65,194	199,419	87,134
#InformationEntropy	5.0193	7.5665	7.0838	3.7947
#MutualInformation	0.0000	0.2577	0.9313	1.3231
#NetworkDensity	0.0949	0.1500	0.0557	0.0017
#ClusteringCoefficient	0.1415	0.0225	0.0534	0.2673
#Heterogeneity	0.0374	0.0076	0.0090	0.0616
#AverageDegreeCentrality	0.0962	0.2541	0.1266	0.0178

5.1. Experiment setups

5.1.1. Datasetss

To confirm the superiority of our algorithm, we conduct a comparison and analysis on four real datasets: FrcSub, JunYi, ASSIT2017 and ASSIT0910. All four public datasets include student response logs and relations between exercises and knowledge concepts. FrcSub includes middle school students' scores on objective exercises on fraction subtraction. JunYi is collected by JunYi platform in Taiwan. ASSIST2017 is a dataset from 2004 to 2007 on the "The 2017 ASSISTments Datamining Competition". ASSIT0910 is a dataset from the 2009–2010 school year collected by the online tutoring system ASSISTments. The train and test sets are divided in a ratio of 8:2. The statistical information of these datasets is presented in Table 2.

5.1.2. Comparison baselines

By comparing eight cognitive diagnostic baselines, we confidently demonstrate the excellent performance and reliability of our proposed method. In addition, we divide the comparison baselines into two types, one is the classical CDM, i.e., IRT (Embretson & Reise, 2013), MIRT (Ackerman et al., 2003) and DINA (De La Torre, 2009). The other is the deep learning-based CDM, i.e., NCDM (Wang et al., 2020), SCD (Shen et al., 2024), KSCD (Ma et al., 2022), KaNCD (Wang et al., 2022), UCD (Wang et al., 2024a), ECFKG (Ai et al., 2018) and KGAT (Wang et al., 2019). Specifically, to validate the innovation of the knowledge graph feature learning module within the CD-SKG, we employ two classic knowledge graph-based feature learning methods, ECFKG and KGAT, to extract entity and relationship features for subsequent cognitive diagnosis.

5.1.3. Evaluation metrics

We evaluate the performance by three evaluation metrics commonly used in the field of cognitive diagnosis: accuracy (ACC), area under the curve (AUC), and root-mean-square error (RMSE). ACC is used to assess the diagnostic accuracy of the model. AUC provides a more comprehensive assessment of the results, which takes into account the diagnostic effect at different thresholds. RMSE measures the average error between the true score and the predicted score.

5.1.4. Parameter settings

We employ Xavier initialization to all parameters in our model. Both P-HGCN and N-HGCN use two layers of HGCN. In Eqs. (9) and (11), σ_1 applies the $LeakyRelu$ activation function with a dropout of 0.5 for the first layer and 0.2 for the second layer. σ_2 applies the Min-Max Scaling activation function to map the values between 0 and 1. The model is trained under the $Adam$ optimizer. All experiments are executed in PyTorch 3.8, 12th Gen Intel(R) Core(TM) i5-12400 2.50 GHz 64 GB memory GPU. We analyze hyperparameters such as embedding size, learning rate, batch size and weight decay in Section 5.3.

5.2. Performance comparison (Q1)

To verify the performance of model, we present the results of our proposed CD-SKG along with all comparison baselines. All the comparison baselines are set up according to the optimal parameters mentioned in their articles and each baseline takes its best result. Our results are bolded and the best baseline results underlined in Table 3. We have the following observations. (1) CD-SKG outperforms the other compared baselines, which demonstrates that our model effectively improves the precision of cognitive diagnostic results. (2) On the FrcSub, our model improves by 1.279%, 2.266%, and 0.864% on each of the three evaluation metrics. Besides, on the ASSIT2017 and ASSIT0910 our model improves by more than two percent on each metric, which indicates that our proposed CD-SKG achieves better performance on both small and larger datasets. (3) The enhancement ratios of our model for the three metrics on JunYi are lower than the ones on the other three datasets. The reason is that each fact in JunYi contains only one knowledge concept, while each fact in the other three datasets contains more than one knowledge concept. More knowledge concepts in a fact helps to explore more higher-order information, which allows for a richer representation of the cognitive state of students. However, if a fact contains only one concept, it cannot capture the higher-order information within student factor and exercise factor, resulting in only second-order information between student and concept or between exercise and concept. Additionally, it cannot explore the extent to which certain concept combinations influence a specific exercise, leading to the conclusion that this concept completely determines the complexity of the exercise. Overall, the superiority of CD-SKG is attributed to that we differentiate different types of responses and enrich the features of the students by exploring higher-order information between different entities.

In addition, to comprehensively verify the effectiveness and superiority of CD-SKG, we further conduct an in-depth analysis of the signed cognitive hypergraphs constructed from different datasets through six properties of information theory and graph theory, and also perform correlation analysis by integrating algorithm performance and structural characteristics. As shown in Table 2: (i) The high information entropy of JunYi and ASSIT2017 indicates that CD-SKG can efficiently capture diverse relationships in signed cognitive hypergraphs and more flexibly model complex cognitive dependencies. (ii) ASSIT0910 has the highest mutual information, directly demonstrating that CD-SKG can maximize the information sharing between student cognition and knowledge concepts in symbolic cognitive hypergraphs, providing high-quality features for subsequent diagnostic tasks; while the mutual information of FrcSub is 0, indicating that CD-SKG does not force linear dependency relationships but models complex associations through different types of response relationships, thereby preserving possibilities for exploring potential cognitive paths in FrcSub. (iii) The high network density of JunYi shows that the nodes in its symbolic cognitive hypergraph are closely connected, and CD-SKG can still maintain efficient computation and low redundancy. (iv) The high clustering coefficient of ASSIT0910 reveals the existence of a large number of triangular structures in its symbolic cognitive hypergraph, providing a natural advantage for CD-SKG to mine diverse high-order structures within facts. (v) The low heterogeneity of JunYi indicates that its node degree distribution is relatively uniform, and in this case, CD-SKGcan avoid the "central node monopoly" problem and ensure that all knowledge concepts can be modeled equally. (vi) The high average degree centrality of JunYi shows that its nodes have a large number of average connections, and CD-SKG optimizes the efficiency of information propagation through the hypergraph structure. Through the above analysis, it can be seen that CD-SKG demonstrates good performance and adaptability when facing datasets with different characteristics.

5.3. Analysis of hyperparameter sensitivity (Q2)

We first analyze the impact of four hyperparameters on the performance of our model in Fig. 5: embedding size d, learning rate lr, batch size $batch$, and weight decay λ. We change each hyperparameter independently and evaluate the performance of the model. Our goal is to determine the optimal hyperparameters to produce the highest accuracy and generalization ability. The results show that the FrcSub is the lowest in sensitivity to hyperparameters. However, the other three larger datasets have more significant sensitivity to hyperparameters. These findings help us to slightly improve our model for better performance. The best hyperparameters on four datasets are presented in Table 4.

Subsequently, we analyze the impact of the feature fusion operation $g(\cdot)$ of the positive embedding H^+ and negative embedding H^- in Eq. (13) on the performance of the model. We select the four solutions "$addition$", "$subtraction$", "$product$" and "$splicing$" for the feature fusion operation, and conduct experiments to compare the effect of these four solutions on different datasets. "$addition$" indicates the addition of two corresponding elements in the embedding, denoted as $H = H^+ + H^-$. "$subtraction$" indicates the subtraction of two corresponding elements in the embedding, denoted as $H = H^+ - H^-$. "$product$" represents the product of two elements at corresponding positions in the embedding, denoted as $H = H^+ * H^-$. Since splicing two embeddings together increases the dimension of the embedding, we adopt a linear transformation function $f(\cdot)$ to reduce the spliced embedding to its original dimension, and the "$splicing$" is denoted as: $H = f(\psi(H^+, H^-))$. Fig. 6 shows that the "$addition$" solution integrates the information between different embeddings and helps model to learn the relation of the data better. Therefore, we adopt the "$addition$" solution as the positive and negative feature fusion operation.

To validate CD-SKG's universality and perturbation resistance, we use the ASSIT2017 dataset, randomly removing 20% of student-exercise or exercise-concept associations to simulate real-world data missing/errors. Compared to KaNCD (optimal baseline under normal conditions), CD-SKG shows a 3.27% ACC and 3.23% AUC

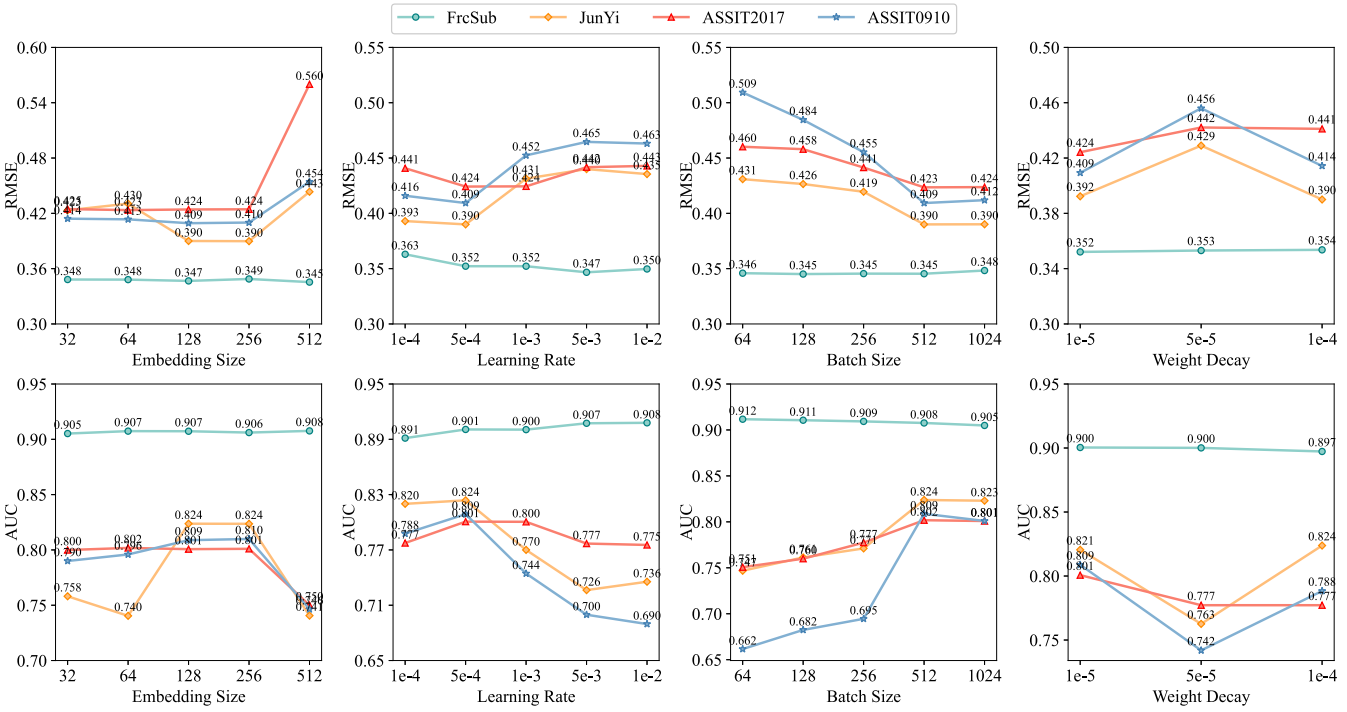


Fig. 5. The impact of hyperparameters.

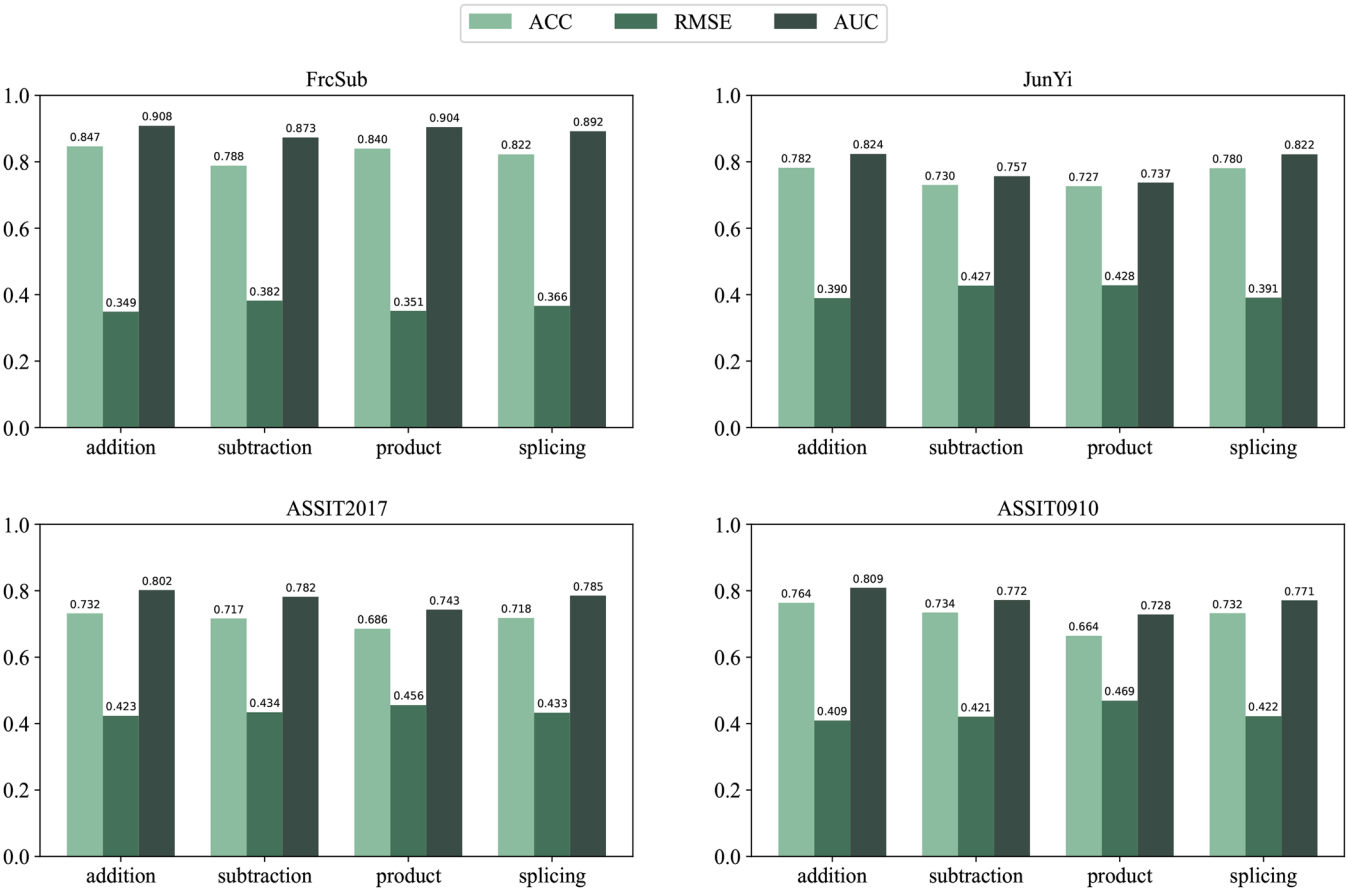


Fig. 6. The impact of feature fusion operations on four datasets.

Table 3
Comparison results of different methods. ↑ indicates that larger value means better result and ↓ indicates the opposite. *Imp.(%)* represents the relative improvement over the suboptimal model.

Methods	FrcSub			JunYi			ASSIT2017			ASSIT0910		
	ACC ↑	RMSE ↓	AUC ↑	ACC ↑	RMSE ↓	AUC ↑	ACC ↑	RMSE ↓	AUC ↑	ACC ↑	RMSE ↓	AUC ↑
IRT	0.5545	0.5506	0.5131	0.7338	0.4214	0.7602	0.6518	0.4646	0.7045	0.6704	0.4671	0.6391
MIRT	0.5569	0.5657	0.5741	0.6826	0.4864	0.6662	0.5454	0.6236	0.5630	0.5840	0.6009	0.5536
DINA	0.7234	0.4495	0.7704	0.6845	0.4622	0.7323	0.6130	0.5120	0.6927	0.6739	0.4781	0.7066
NCDM	0.8225	0.3951	0.8842	0.7623	0.4064	0.8004	0.6958	0.4448	0.7588	0.7356	0.4323	0.7598
SCDM	0.8183	0.3680	0.8935	0.7532	0.4830	0.7716	0.6921	0.5267	0.7378	0.7283	0.4615	0.7375
KSCD	0.8267	0.3531	0.9028	0.7725	0.3951	0.8205	0.7029	0.4402	0.7785	0.7306	0.4242	0.7637
KaNCD	0.8324	0.3653	0.8940	0.7774	0.3928	0.8218	0.7125	0.4355	0.7790	0.7331	0.4313	0.7649
UCD	0.8330	0.3679	0.8962	0.7711	0.4013	0.8138	0.7075	0.4417	0.7737	0.7375	0.4290	0.7667
ECFKG	0.8288	0.3605	0.8979	0.7793	0.3916	0.8213	0.7269	0.4278	0.7956	0.7369	0.4211	0.7726
KGAT	0.8368	0.3556	0.8993	0.7814	0.3910	0.8226	0.7301	0.4243	0.7997	0.7318	0.4234	0.7664
CD-SKG	**0.8475**	**0.3451**	**0.9106**	**0.7825**	**0.3900**	**0.8238**	**0.7318**	**0.4233**	**0.8019**	**0.7636**	**0.4093**	**0.8088**
Imp.(%)	1.279%	2.266%	0.864%	0.141%	0.256%	0.146%	0.233%	0.236%	0.275%	3.539%	2.802%	4.685%

Table 4
The best hyperparameter values for different datasets.

Datasets	*d*	*lr*	*batch*	*λ*
FrcSub	512	5e−3	128	1e−5
JunYi	128	5e−4	512	1e−4
ASSIT2017	128	5e−4	512	1e−5
ASSIT0910	128	5e−4	512	1e−5

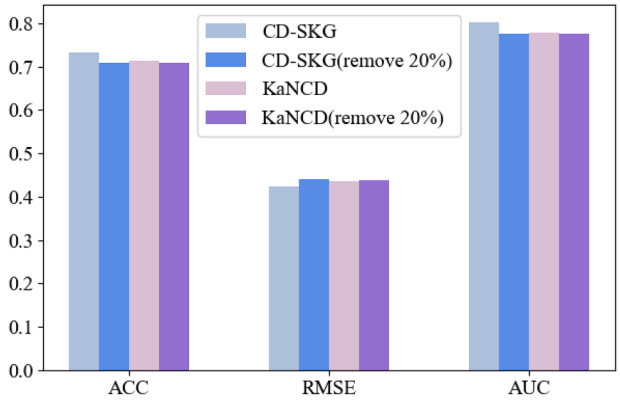


Fig. 7. The impact of feature fusion operations on four datasets.

decline post-perturbation, while KaNCD degrades minimally (0.66% ACC, 0.29% AUC). Despite this sensitivity, CD-SKG's perturbed metrics still exceed KaNCD's undisturbed performance (by 2.1% ACC, 1.8% AUC). RMSE increases more for CD-SKG (4.02%) than KaNCD (0.55%), but remains acceptable. CD-SKG outperforms KaNCD in high-quality data scenarios (5.2% ACC, 4.1% AUC baseline) but exhibits vulnerability to structural perturbations. It is best suited for environments with robust association completeness; data imputation is recommended for high-missingness contexts. This balance of superior baseline performance and perturbation sensitivity defines its operational scope, emphasizing perturbation resistance over generic robustness in multi-scenario applications.

5.4. Study of ablation (Q3)

As shown in Table 5, we perform ablation experiments on three modules. $\mathcal{G}$ represents the signed cognitive hypergraph. To validate the effectiveness of $\mathcal{G}$, the correlation matrix between entities and relations within $\mathcal{G}$ are removed. Then $\mathcal{G}$ is replaced with a simple heterogeneous graph using students' correct or incorrect responses to exercises as edge types. $\mathcal{F}_{ij}^{CD}$ represents the cognitive factors obtained at the concept-aware diagnosis level. $\mathcal{F}_{ij}^{KG}$ represents the combination of two cognitive factors obtained at the knowledge graph diagnosis level. We explore the impact of the three modules in turn, and later explore the joint impact of the $\mathcal{F}_{ij}^{KG}$ and $\mathcal{F}_{ij}^{CD}$ modules. We have the following findings. (i) The experimental results show that removing any of the components from the model degrades the performance of the model, which indicates that each component plays a significant role. (ii) $\mathcal{G}$ ensures that the features of entities and relations learned in dual-view HGCN are distinct for different types of responses while avoiding interference from positive and negative cognitive features. Removing $\mathcal{G}$ resulted in an average 3.2% decrease in ACC across four datasets, particularly a 3.22% decline on ASSIT0910. This significant difference demonstrates that the signed cognitive hypergraph effectively captures cognitive information that traditional heterogeneous graphs cannot adequately represent. (iii) $\mathcal{F}_{ij}^{CD}$ is a cognitive factor with higher-order information obtained through a simple student-exercise interaction function, and it has a small effect on JunYi. This is due to that each exercise contains only one knowledge concept in JunYi, so $\mathcal{F}_{ij}^{CD}$ is difficult to explore higher-order concept-aware structures centred on students or exercises. (iv) $\mathcal{F}_{ij}^{KG}$ ensures the original features of the individual entities within the fact as well as the higher-order features between different entities, and it has a significant effect on four datasets. (v) The combination of $\mathcal{F}_{ij}^{KG}$ and $\mathcal{F}_{ij}^{CD}$ has the most significant effect on four datasets, which indicates that the various higher-order information exists in the facts effectively improve the cognitive state of students. The above observations further verify the efficiency of introducing signed cognitive hypergraph and two diagnosis levels.

5.5. Analysis of the signed cognitive diagnosis (Q4)

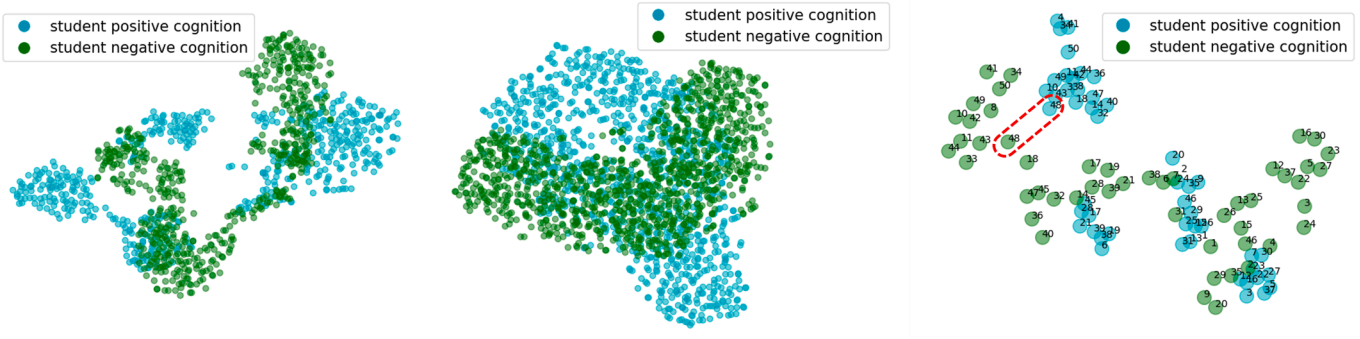
To distinguish the ability of student in different response data, we construct two subgraphs: the positive cognitive hypergraph and the negative cognitive hypergraph, as detailed in Section 4.1. Here, we downscale and visualize the two embeddings of students through *t*-SNE (Van der Maaten & Hinton, 2008) to show the influence of signed cognitive diagnosis. The *t*-SNE (*t*-distributed Stochastic Neighbor Embedding) algorithm is a nonlinear downscaling and data visualisation technique for mapping high-dimensional data into a two or three dimensional space while maintaining local relations between nodes. As shown in Fig. 8, we clearly exhibit the two embeddings for all students in two colors on FrcSub and JunYi. We observe that (i) there is a significant difference in the distribution of the two embeddings, which reflects the difference between students in terms of positive and negative cognition; (ii) there is a noticeable difference in the two types of embeddings, which further indicates certain similarity between students with the same cognition; (iii) there are also overlapping parts between the two types of embeddings, which is possible that cognitions of different students overlap. In addition to the student-exercise interactions are extremely intricate, there are a number of other potential factors (i.e., guess and slip) that affect the cognitive abilities. The above phenomena collectively reflect the complexity of students' cognitive states, more specifically revealing that the same student may exhibit distinct cognitive patterns under different response types. In simple heterogeneous graphs, correct or incorrect responses exist solely as edge attributes, incapable of forming independent topological structures. This demonstrates the necessity of proposing the symbolic cognitive hypergraph.

We further display 50 randomly selected students on the FrcSub in Fig. 8(c), with different labels representing different students. For example, the two embeddings of the 48th student are not close to each other, which illustrates different personal cognitive abilities for different response data. This provides us with powerful support for subsequent in-depth analyses of the cognitive states. At the same time, this individualized cognitive state also provides us with an intuitive way to understand the learning cognitions of students.

We observe distinct distributions between two embeddings, reflecting students' differences in positive and negative cognition, with noticeable disparities indicating similarity among students sharing the same cognition and overlaps

Table 5
Ablation study.

Methods			FrcSub			JunYi			ASSIT2017			ASSIT0910		
$\mathcal{G}$	$\mathcal{F}_{ij}^{CD}$	$\mathcal{F}_{ij}^{KG}$	ACC ↑	RMSE ↓	AUC ↑	ACC ↑	RMSE ↓	AUC ↑	ACC ↑	RMSE ↓	AUC ↑	ACC ↑	RMSE ↓	AUC ↑
✗	✓	✓	0.8036	0.3760	0.8855	0.7804	0.3908	0.8223	0.7159	0.4342	0.7843	0.7314	0.4236	0.7678
✓	✗	✓	0.8204	0.3726	0.8905	0.7810	0.3901	0.8234	0.7112	0.4383	0.7775	0.7361	0.4216	0.7731
✓	✓	✗	0.8344	0.3551	0.8979	0.7737	0.3980	0.8119	0.7157	0.4357	0.7820	0.7355	0.4234	0.7706
✓	✗	✗	0.7938	0.3853	0.8805	0.7671	0.4020	0.8055	0.7134	0.4390	0.7765	0.7259	0.4281	0.7588
✓	✓	✓	**0.8475**	**0.3451**	**0.9106**	**0.7825**	**0.3900**	**0.8238**	**0.7318**	**0.4233**	**0.8019**	**0.7636**	**0.4093**	**0.8088**



(a) Student embeddings in FrcSub. (b) Student embeddings in JunYi. (c) 50 student embeddings in FrcSub.

Fig. 8. Positive and negative cognitive distributions.

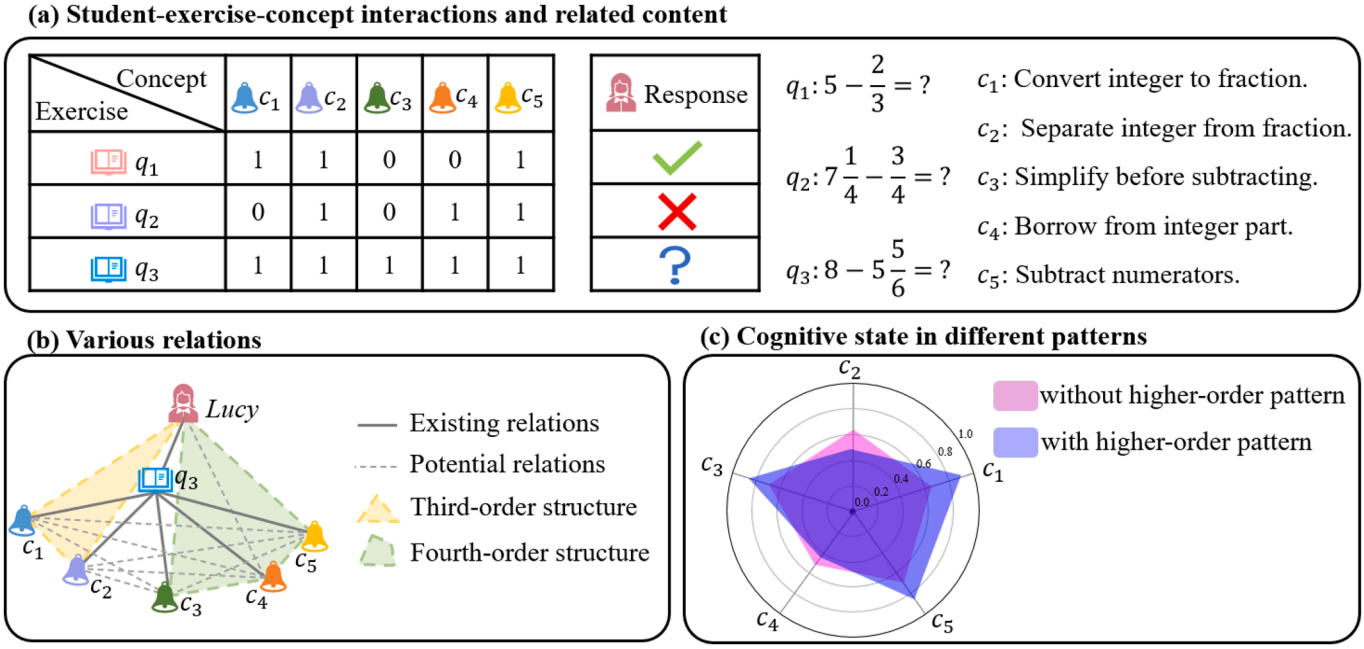


Fig. 9. Illustrative examples of (a) response of *Lucy* and related knowledge concepts; (b) various relations between entities and higher-order structures within the predicted fact $t_3 = r_p(Lucy, q_3, c_1, c_2, c_3, c_4, c_5)$; (c) visualizing the cognitive states of *Lucy* in two patterns.

suggesting potential cognitive overlaps between different students. Embedding differences support in-depth cognitive state analysis and offer an intuitive understanding of students' learning cognitions–highlighting the importance of modeling positive and negative facts.

5.6. Interpretation of the diagnosis (Q5)

To verify the interpretability of our proposed CD-SKG, we select the response log of *Lucy* on FrcSub mentioned in Fig. 1, and then visualize the knowledge concepts mastery of *Lucy* with/without incorporating higher-order structural information, respectively. As shown in Fig. 9, we obtain the cognitive state without higher-order information from the history responses of *Lucy*, and then extract the higher-order structural information within the predicted fact $t_3 = r_p(Lucy, q_3, c_1, c_2, c_3, c_4, c_5)$ to obtain the higher-order cognitive state. These two cognitive states ultimately combine to predict the probability of fact t which means the score of *Lucy* on the exercise q_3. Fig. 9(a) illustrates *Lucy*'s responses in exercises q_1 and q_2, along with the corresponding knowledge concepts and a detailed description of these concepts. Fig. 9(b) describes various relations between entities and higher-order structures within the predicted fact t_3. For example, the second-order relationship between c_2 and c_5, and the third-order relationship between Lucy and both c_2 and c_5. These structures reveal how knowledge concepts interact to influence student performance. For instance, when exercise q_3 involves all concepts from c_1 to c_5, Lucy's mastery of c_2 depends not only on c_2 itself but

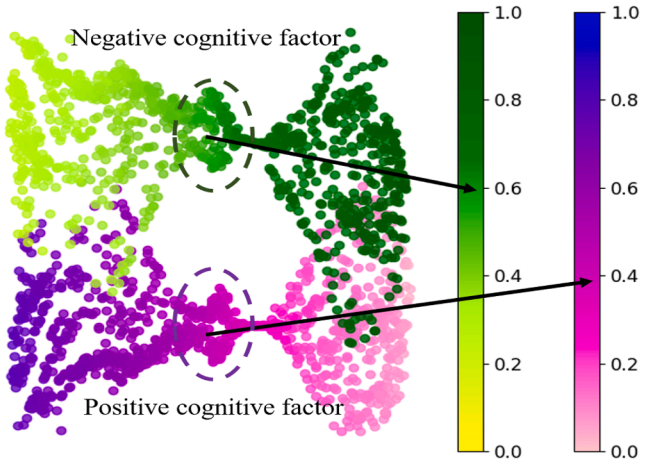


Fig. 10. Illustrative examples of cognitive factors $\mathcal{F}_{ij}^{-}$ and $\mathcal{F}_{ij}^{+}$ on FrcSub.

also on how c_2 interacts with other knowledge concepts. From Fig. 9(c), it can be observed that prior to integrating higher-order structural information. *Lucy* has relatively high mastery of knowledge concepts c_1, c_2, c_3 and c_5, and relatively low mastery of knowledge concept c_4. After incorporating higher-order structural information, student mastery of knowledge concepts c_1, c_3 and c_5 increases significantly, while mastery of knowledge concepts c_2 and c_4 decreases. The finding further demonstrates the efficiency and interpretability of our exploration of higher-order information within facts. Specifically, Lucy's mastery of c_2 drops from 0.65 to 0.48 after the integration of higher-order information. This happens because the model detects a negative interaction between c_2 and c_4 in Lucy's response to q_2, which most likely stems from conceptual confusion. Consequently, Lucy takes the interactions among different knowledge concepts into full account when answering q_3. This leads to more realistic prediction outcomes. In fact, Lucy answers q_3 correctly with consistent prediction results, confirming that higher-order cognitive state can improve the primitive cognitive state. These higher-order insights make diagnostic results actionable for educational guidance. The higher-order information captured by the model essentially reflects the interactive structural patterns of knowledge concepts in educational scenarios. These patterns show that students do not master concepts in isolation; instead, they form interconnected networks of conceptual relationships. By modeling these higher-order structures, our model advances diagnostic insights from "what" (whether students master concepts) to "why" (the conditions under which mastery succeeds or fails). This provides a scientific basis for precise educational interventions.

Subsequently, to illustrate the efficiency of the cognitive factors, we analyze them in a visualized experiment. We utilize the cognitive factors $\mathcal{F}_{ij}^{-}$ and $\mathcal{F}_{ij}^{+}$ derived from knowledge graph diagnosis level to scale them down to a two-dimensional plane through t-SNE for node visualization. These factors are then mapped onto the knowledge conceptual space using a linear transformation function and normalized. Each dimension is subsequently treated individually as a measure of mastery regarding specific knowledge concepts for each of the two cognitive factors. As shown in Fig. 10, color bars are employed to distinguish the degree of knowledge concept mastery. The left color bar represents the mastery level of negative cognitive factors, with intensity increasing from bottom to top; similarly, the right color bar indicates the mastery level of positive cognitive factors, with intensity also increasing upward. Our observations reveal significant variability in how the two cognitive factors master specific knowledge concepts. Furthermore, we select two regions along the same vertical line for each cognitive factor. A clear contrast in color intensity is evident between these regions, indicating that the cognitive levels of the two factors exhibit complementary characteristics within the same dimension. This complementarity cannot be effectively modeled using edge type attributes within a single graph structure, as information propagation would interfere with one another, and thus demonstrates the necessity of proposing the signed cognitive hypergraph.

6. Conclusion

In this paper, we propose a cognitive diagnosis framework based on signed knowledge graph with multi-fact reasoning. First, in terms of exploring complex relations between entities, we model the interactions among students, exercises and knowledge concepts as a signed cognitive knowledge hypergraph. At the same time, each response log is considered as a signed fact helps us to obtain a comprehensive picture of student learning and performance. Second, to distinguish the impact of different response types on student performance, we use a dual-view HGCN. By learning the features of different responses separately on a signed cognitive hypergraph, we are able to accurately analyze their impact on student performance. Finally, higher-order information on the three cognitive factors are integrated and analyzed at two diagnosis levels to explore the different higher-order relations in each response record. Extensive experiments on four datasets demonstrate that our proposed CD-SKG outperforms state-of-the-art baselines, indicating that dual-view HGCNs and exploring higher-order information among student-exercise-concept are meaningful for cognitive diagnosis.

However, the acquisition of the cognitive state of the student is influenced in various ways due to the complexity of student interactions. For example, there are hierarchies between knowledge concepts, which makes the cognitive state of students for the latter concepts influenced by the former concepts (Liu et al., 2023). Secondly, the cognitive state of students is also influenced by other students within the group (Yu et al., 2024). Therefore, we will further explore and optimize the hierarchical relations and grouping relations between different higher-order structures to better analyze the cognitive states of students in the future. Meanwhile, we plan to explore robust mechanisms(e.g., graph denoising modules or adaptive filtering strategies) for handling noisy graph data in our future research.

CRediT authorship contribution statement

Chengfeng Liu: Writing – original draft, Validation; **Jianrui Chen:** Writing – review & editing, Methodology; **Zhihui Wang:** Visualization, Data curation; **Longjiang Guo:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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